

UET University Peshawar

Abbottabad Campus



Lab Report No: 06

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Class No: 21

Course: SE-305L Computer Communications and Networks

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Lab 6: Visualizing Network Traffic Using Simulation Mode in Packet Tracer

Objective

To introduce students to Simulation Mode in Packet Tracer to visualize how data (packets) flows through different network topologies and observe differences in behavior.

Learning Outcomes

By the end of this lab, students will be able to:

- Use Simulation Mode in Packet Tracer Observe packet movement between devices (e.g., ICMP, TCP) Compare traffic behavior in Star vs Mesh vs Bus topologies Understand broadcasts, collisions, hops, and latency differences Visually compare how different topologies handle traffic flow

Part A: Star Topology with Simulation Mode Step

1: Create Star Topology

- Drag 1 Switch and 4 PCs Connect each PC to the Switch using Copper Straight-Through cables

Step 2: Assign IP Addresses

PC	IP Address	Subnet Mask
PC0	192.168.10.1	255.255.255.0
PC1	192.168.10.2	255.255.255.0
PC2	192.168.10.3	255.255.255.0
PC3	192.168.10.4	255.255.255.0

Click each PC > Desktop > IP Configuration

Step 3: Switch to Simulation Mode

- On the bottom-right of Packet Tracer, change from Realtime to Simulation

Step 4: Generate Traffic

- Click Add Simple PDU Tool (**Envelop** option)

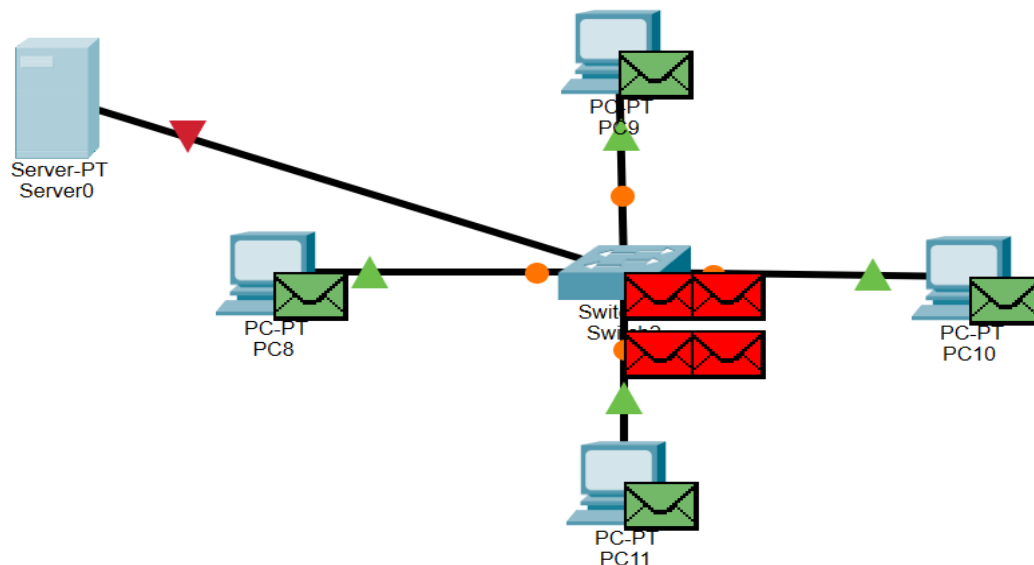
Click PC0, then PC2

Do another one from PC1 to PC3

Step 5: Play the Simulation

- Click Capture / Forward repeatedly to step through the packet journey Observe packet type: ICMP (Ping), direction, device, success/failure

Simulation covering all previous steps.



Part B: Add a Web Server for HTTP Traffic

Step 1: Add Server

Place 1 Generic Server.

Assign IP: 192.168.10.100

Services tab: Ensure HTTP is ON

Step 2: Connect the server with the switch

Step 3: Use Web Browser from any PC

PC > Desktop > Web Browser > type 192.168.10.100

In Simulation Mode, this shows TCP + HTTP packets

Simulation Panel					
Event List					
Vis.	Time(sec)	Last Device	At Device	Type	
	10.099	--	Switch2	CDP	
	10.099	--	Switch2	CDP	
	10.099	--	Switch2	CDP	
	10.100	Switch2	PC9	CDP	
	10.100	Switch2	PC10	CDP	
	10.100	Switch2	PC11	CDP	
	10.100	Switch2	PC8	CDP	
	12.096	--	Switch2	STP	
	12.097	Switch2	PC10	STP	
	12.097	Switch2	PC8	STP	
	12.097	Switch2	PC9	STP	
	12.097	Switch2	PC11	STP	

Reset Simulation ☒ Constant Delay Captured to: 12.097 s

Part C: Mesh Topology Simulation

Step 1: Create Mesh Topology Using Switches

- Place 4 PCs and 4 Switches on the screen.

Connect each PC to a dedicated Switch.

Interconnect the switches in full mesh using straight-through cables.

Step 2: Assign IP Addresses

PC	IP Address	Subnet Mask
PC0	192.168.20.1	255.255.255.0
PC1	192.168.20.2	255.255.255.0
PC2	192.168.20.3	255.255.255.0
PC3	192.168.20.4	255.255.255.0

Step 3: Simulation Traffic

- Use Add Simple PDU Tool again.

PC0 to PC1

PC2 to PC3

Step 4: Play the Simulation

Use Capture / Forward to step through the packet journey.

Step 5: Add a Web Server for HTTP Traffic

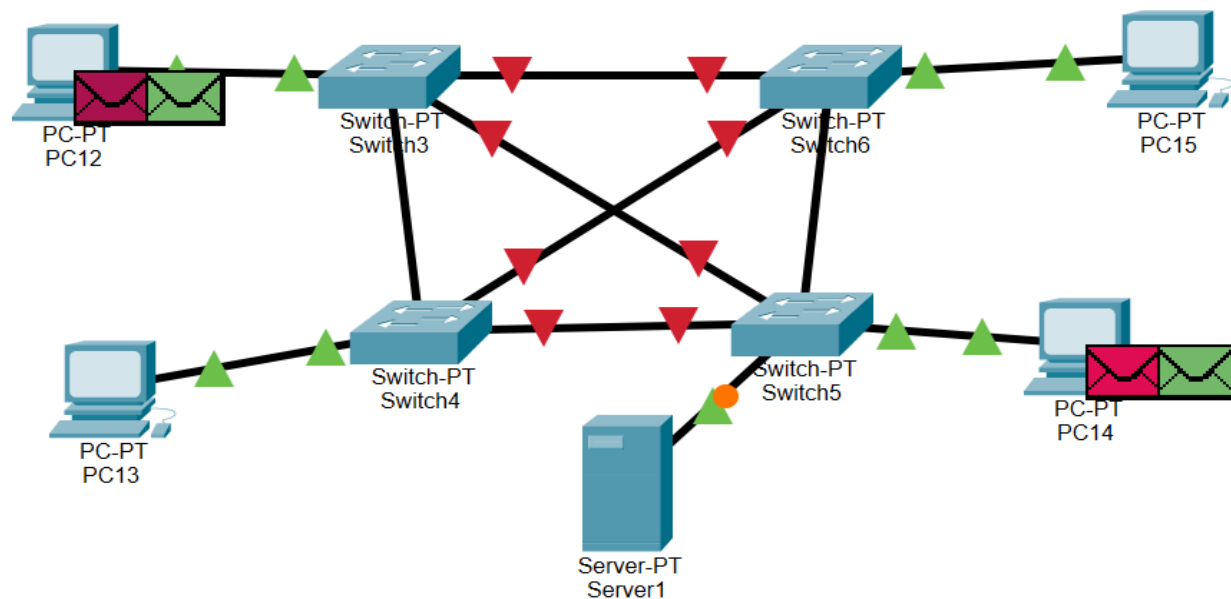


Figure 1 Simulation

Event List				
Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC12	ICMP
	0.000	--	PC12	ARP
	0.000	--	PC14	ICMP
	0.000	--	PC14	ARP

Part D: Bus Topology Simulation

Step 1: Create Bus Topology

Drag 4 PCs and 1 Hub onto the workspace.

Connect each PC to the Hub using Copper Straight-Through cables.

Step 2: Assign IP Addresses

PC	IP Address	Subnet Mask
PC0	192.168.30.1	255.255.255.0
PC1	192.168.30.2	255.255.255.0
PC2	192.168.30.3	255.255.255.0
PC3	192.168.30.4	255.255.255.0

Step 3: Generate Traffic

Step 4: Play the Simulation

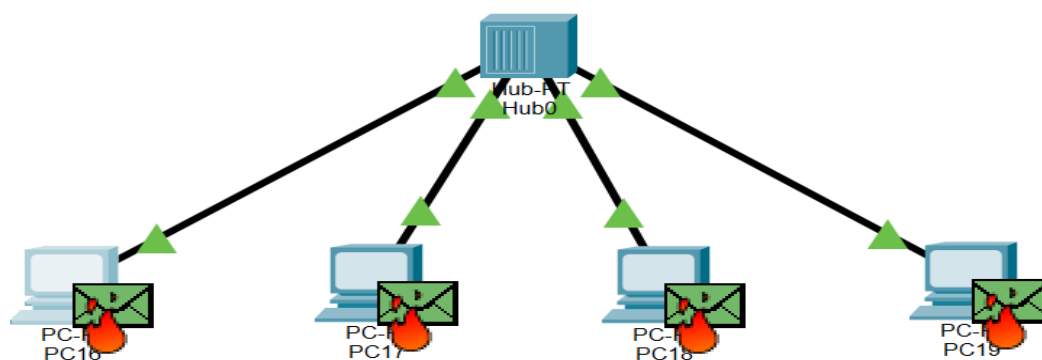
Observation Points All packets broadcast to all ports (hub has no intelligence).

Multiple packets may collide or interfere (symbolized by "X" marks).

There's no filtering or routing of traffic.

Notice how the hub floods all packets to all connected PCs.

Step 5: Add HTTP Traffic



Part E: Comparison Table

Feature: Number of Hops

Star Topology_____1_____

Mesh Topology:_____4_____

Bus Topology:_____1_____

Feature: Broadcast Behavior

Star Topology:_____Broadcast Flooding_____

Mesh Topology:_____Perfect broadcasting_____

Bus Topology:_____All devices receive every broadcast _____

Feature: Visual Clarity

Star Topology:_____Yes_____

Mesh Topology:_____No_____

Bus Topology:_____Yes_____

Feature: Cabling Cost

Star Topology:_____Medium Cost_____

Mesh Topology: _____High Cost_____

Bus Topology: _____Low Cost_____

Feature: Collisions

Star Topology:_____No_____

Mesh Topology: _____No_____

Bus Topology: _____YES(In case of simultaneous transmission_____

That's it!